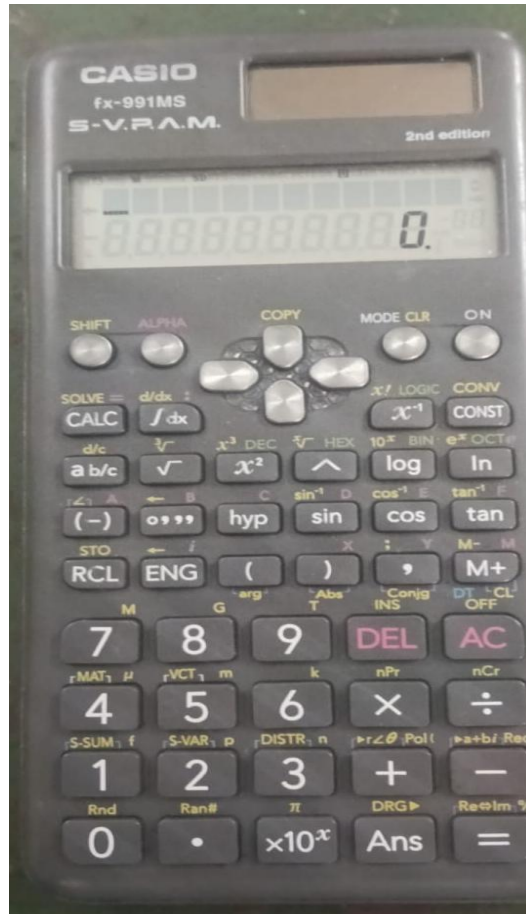




Scientific Calculator: Inside & Out

Casio fx-82MS: Embedded Systems

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What is a Scientific Calculator?

A scientific calculator is an electronic device designed for complex mathematical, scientific, and engineering computations. Unlike basic calculators, it performs advanced functions like trigonometry, logarithms, and statistics.

Key Applications

Essential for engineering, scientific research, and advanced academic studies.



Overall Internal Architecture

The Casio fx-82MS features a modular design, typical of embedded systems.



01

Input Module

Keypad captures user commands.

03

Output Module

LCD displays results.

02

Processing Unit

Microcontroller runs algorithms.

04

Power System

Battery and voltage regulation.

Main Parts

- LCD Display
- Printed Circuit Board (PCB)
- Microcontroller (ASIC Chip)
- Keypad (Rubber Membrane + Plastic Keys)
- Battery (AAA)
- Outer Casing
- Battery Contacts & Screws

Materials Used

Outer Case & Keys: ABS Plastic

PCB: Fiberglass (FR4) with Copper traces

Microcontroller Chip: Silicon (covered with Epoxy resin)

LCD: Glass + Liquid Crystal + Indium Tin Oxide

Keypad Membrane: Rubber with Conductive Carbon pads

Battery Contacts: Nickel-plated Steel / Copper alloy

Solder: Lead-free Tin alloy

LCD Display Module

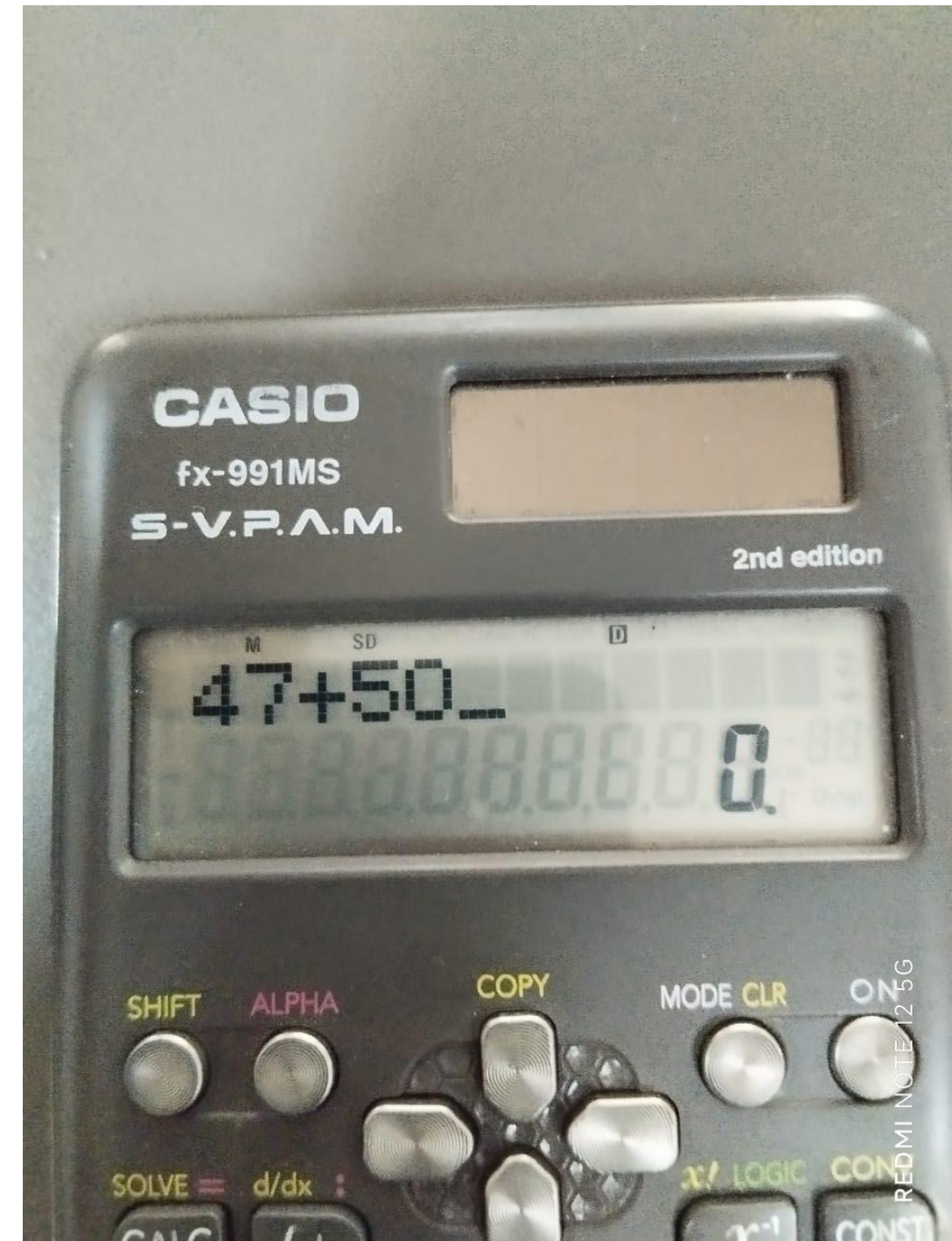
Segmented LCD Technology

Uses fixed segments of liquid crystals to form digits and symbols.

Applying voltage to segments aligns crystals, manipulating light to create visible patterns.

Key Benefits

- **Ultra-low power** (ideal for battery devices)
- **No backlight needed** (excellent daylight visibility)
- **Long operational lifespan**



Printed Circuit Board (PCB)



Base Material

FR4 fiberglass for strength & insulation.



Copper Traces

Pathways connect components & route signals.



Single-Layer Design

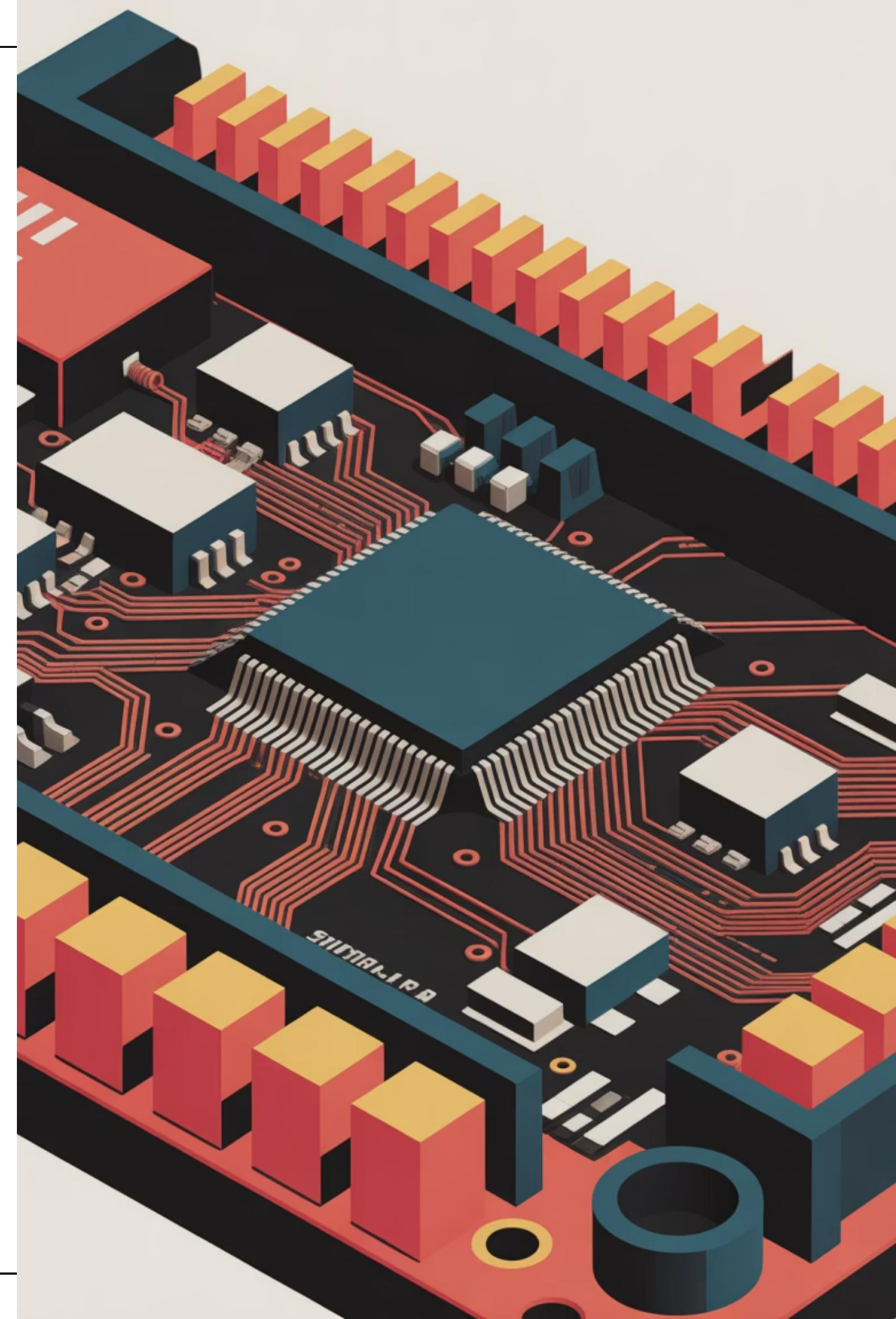
Simple structure with plated through-holes.



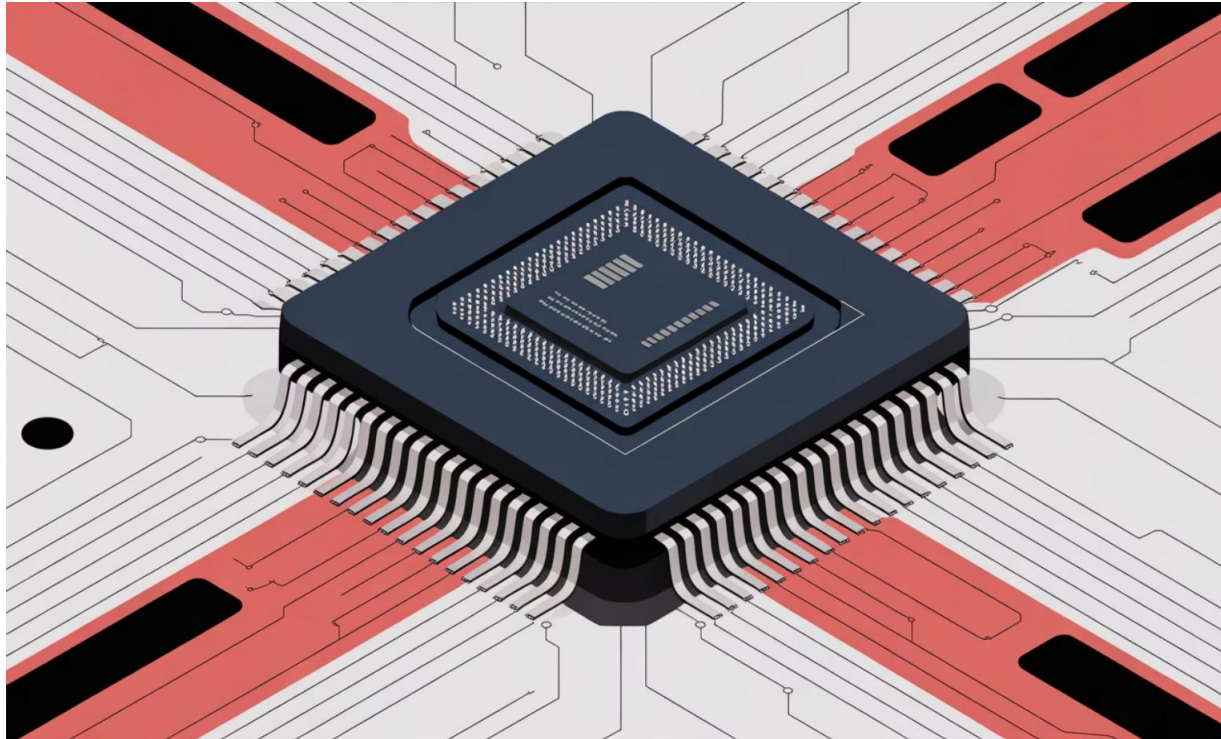
Signal Routing

Traces minimize interference and crosstalk.

The PCB forms the core, integrating all electronic components. Its robust design ensures reliability.



Microcontroller: The Brain



Integrated Circuit Architecture

Chip-on-Board (COB) design, bonded to PCB with gold wires. Compact and cost-effective.

Arithmetic Logic Unit (ALU)

Handles math and logic operations.

Control Unit

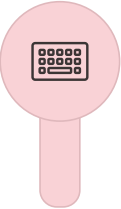
Fetches and decodes instructions.

Memory System

ROM for firmware, RAM for variables.

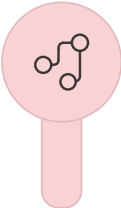
❏ **Firmware:** Calculator's operating software stored in ROM, includes algorithms for all functions.

Keypad System



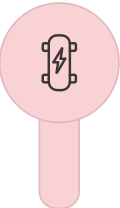
Rubber Membrane

Flexible dome with carbon contact pads



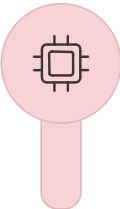
PCB Matrix

Copper traces in rows and columns



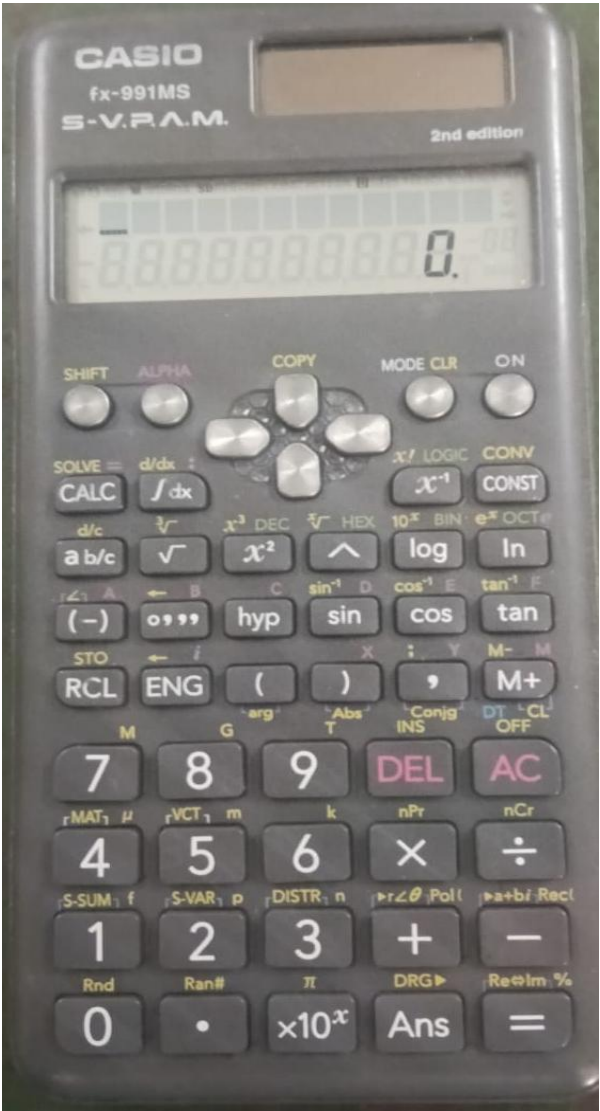
Matrix Scanning

Microcontroller activates rows, reads columns



Key Detection

Pressed key bridges connection, signal detected



Process Flow:

Keypad Matrix Detects Row–Column Signal



Microcontroller Decodes Function



Mathematical Algorithm Executed



Result Stored in RAM & LCD Segments Activated

Power System

1. Power Source

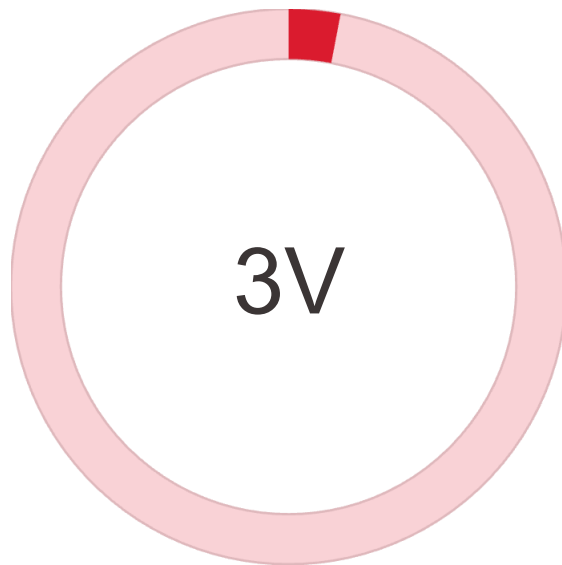
Powered by a single AAA battery (1.5V) or hybrid solar cell for extended life.

2. Voltage Regulation

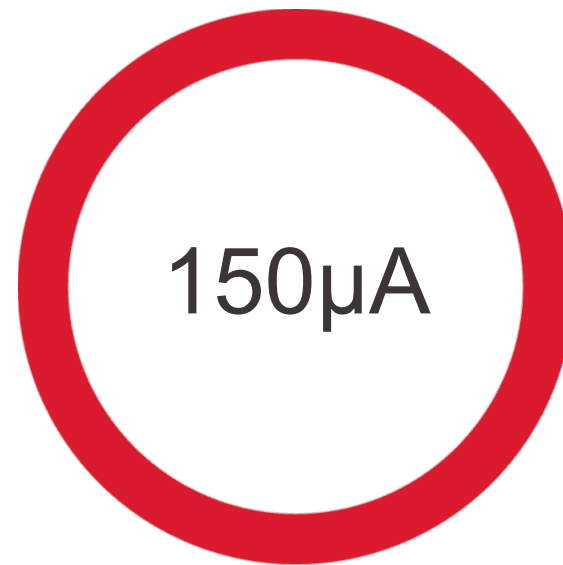
Internal circuits ensure a stable 3V supply for CMOS logic, maintaining reliable operation.

3. CMOS Technology

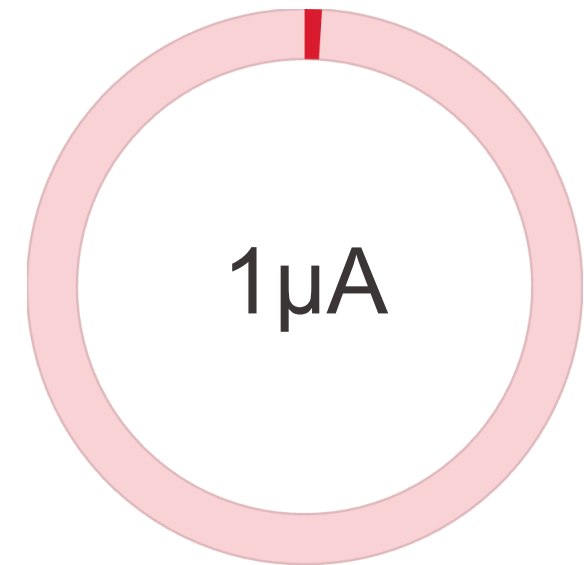
. Utilizes CMOS technology for minimal power consumption, especially in standby mode



Operating Voltage



Active Current



Standby Current

Embedded System Characteristics

Why a Calculator is an Embedded System

- **Modular Architecture:** Efficient computation
- **CMOS Technology:** Power efficiency
- **Dedicated Firmware:** Mathematical algorithms
- **Elegant Design:** Core embedded principles

Dedicated Function
Specific for calculations

Real-time Operation
Instant input/output

Low Power Design
Long battery life

Resource-Constrained
Limited power & memory

Integrated Hardware
Custom firmware optimized



Insight: The Casio fx-82MS showcases reliable, cost-effective embedded solutions.

THANK YOU